Hang Du

Ph.D. Candidate in Software Engineering, University of California, Irvine

hdu5@uci.edu · +1 (949) 537-1188 · superhangdu.com · Google Scholar · LinkedIn

Education

University of California, Irvine 2022 – Present (Expected 2028)

Ph.D. in Software Engineering

Advisor: James A. Jones

University of California, Irvine 2021 – 2025

Master of Science in Software Engineering

Advisor: James A. Jones (Transferred to Ph.D. program Sep 2022; conferred Dec 2025)

Coursework: Program Analysis, Software Architecture, Software Testing, Information Retrieval

McGill University, Montreal, Canada Feb 2020

Machine Learning Certificate Program

Coursework: Computational Applied Statistics, Practical Machine Learning

Jilin University, China 2017 – 2021

Bachelor of Science in Computer Science and Technology

Bachelor of Finance (minor)

Peer-Reviewed Publications

[1] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2026. *How Does Killing Surviving Mutants Help Detect Real Bugs with Assertion Generation: A Controlled Experiment*. In Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026) [PDF].

ACM SIGSOFT Distinguished Paper Award

Direct Acceptance Rate: 10.1%

[2] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2026. *Revealing Regressions: A Comparative Study of State-Capture Strategies in Validating Program Behavior*. Proceedings of the ACM on Software Engineering (FSE 2026); also in the journal Proceedings of the ACM on Software Engineering (PACMSE) [PDF].

Direct Acceptance Rate: 9.5%

[3] Monil Narang, **Hang Du**, and James A. Jones. 2025. *What's DAT Smell? Untangling and Weaving the Disjoint Assertion Tangle Test Smell*. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025) [PDF].

Acceptance Rate: 20.6%

[4] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2025. *Leveraging Propagated Infection to Crossfire Mutants*. In Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE 2025) [PDF].

Oral presentation in Ottawa, Canada (presented by Hang Du). Acceptance Rate: 21.4%

[5] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2024. *Ripples of Mutation: An Empirical Study of Propagation Effects in Mutation Testing*. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE 2024) [PDF].

Oral presentation in Lisbon, Portugal (presented by Hang Du). Acceptance Rate: 21.2%

[6] **Hang Du**, Vijay Krishna Palepu, and James A. Jones. 2023. *To Kill a Mutant: An Empirical Study of Mutation Testing Kills*. In Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2023) [PDF].

Oral presentation in Seattle, USA (presented by Hang Du). Acceptance Rate: 25.9%

Service

Shadow Program Committee Member, 49th IEEE/ACM International Conference on Software Engineering (ICSE 2027) — 2027

Reviewer, ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), Artifact Evaluation Track — 2026

Reviewer, IEEE/ACM International Conference on Automated Software Engineering (ASE), Artifact Evaluation Track — 2026

Reviewer, IEEE Transactions on Software Engineering (TSE) — 2026

Student Volunteer, ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA) — 2026

Junior Program Committee Member, International Conference on Mining Software Repositories (MSR) — 2026, 2027

- Reviewed and evaluated research papers submitted to the MSR Research/Technical Track.
- Provided constructive feedback to authors and participated in the peer-review and discussion process.

External Reviewer, IEEE Transactions on Reliability — 2024

- Reviewed and provided detailed feedback on submitted manuscripts in software reliability and testing.

Graduate Research Mentor, University of California, Irvine — Oct 2023 – Jun 2024

- Mentored an undergraduate UROP (Undergraduate Research Opportunities Program) participant in the development of an honors thesis titled “Generative AI for Mutation Testing.”
- Advised on research topic selection, proposal development for UROP funding, and the creation of a research poster for the 2024 Undergraduate Research Symposium.
- Conducted weekly research meetings to provide guidance and feedback.

Selected Projects

Terminology: a “mutant” is a copy of a program with one artificially injected fake bug, and “mutation testing” seeds many such bugs to check whether the test suite catches them; all uses below carry that meaning.

A Decade of Mutation Testing in the Wild: Practice, Mindset, and Community (ongoing)

- Studying how open-source projects adopt and sustain mutation testing over ten years, characterizing the **testing mindsets and community practices** that form around artificially injected bugs.
- Analyzing the full mutation-testing history of long-running projects such as **Checkstyle**, which has practiced mutation testing for **a decade** and accumulated **over 36,000 artificially injected bugs**.
- Tracing how the practice is evolving: recent projects such as n8n now **delegate mutation testing to AI agents** and enforce **mutation score as a quality-assurance gate** for both human- and AI-written code.
- Deriving insights into how developers reason about **test adequacy**, which mutants they choose to address, and how these decisions evolve as projects mature.

Test Suite Improvement with Assertion Amplification and Mutant Killing

- Proposed a **test-failure taxonomy** of the runtime symptoms through which tests detect mutants and real bugs, and built a **Java analysis framework** that traces how a fault **progressively infects program state and propagates** to a failure; evaluated at scale on artificially injected faults and real bugs from

Defects4J (ISSTA 2023, ICSE 2024).

- Turned those propagation insights into an automated approach that generates **assertion candidates** for human-written test suites and optimizes them to **"crossfire" multiple hidden faults** at once, detecting all detectable faults with only **1.1% of candidates** and boosting fault detection by **6**× on average across ten open-source Java projects and 1.2 million test executions; supporting theoretical model and artifact released (ICSE 2025).
- Ran a **controlled experiment** on whether **killing surviving mutants** actually helps detect **unseen real bugs**: of 642 Defects4J bugs, **104** could have been caught by **one localized assertion** added to an existing passing test (13,266 validated augmentation options generated), and 63 of those were reachable by killing Major mutants with the technique.
- Showed that the benefit does not require exhaustive killing—randomly killing **30%** of relevant surviving mutants, one assertion each, detects **84.5%** of those bugs—and explained the **mechanism**: mutation-derived assertions validate program outputs that **overlap with, and are often subsets of**, the outputs corrupted by real bugs, while real bugs corrupt larger program state than mutants (ISSTA 2026, **ACM SIGSOFT Distinguished Paper Award**).

Technical Skills

Program instrumentation & dynamic analysis: Java bytecode instrumentation, runtime state capture and profiling, fault-propagation tracing, mutation analysis pipelines at scale (1.2M+ test executions), Defects4J

AST-based code transformation: static analysis and source-to-source refactoring with JavaParser, automated test-smell detection and repair, assertion synthesis and amplification

Software testing: JUnit, mocking/stubbing and testable design, test adequacy and mutation scoring, large-scale experiment harnesses, controlled experiments and developer studies

AI for software engineering: LLM-augmented developer tools, AI agents for testing and code review, evaluation of generated tests and code, Model Context Protocol (MCP)

Languages & tooling: Java (primary), Python; Git, Docker, Linux, continuous integration

Teaching & Mentoring

DTEI Summer Graduate Scholar, University of California, Irvine Jul 2026 – Sep 2026

- Awarded a competitive $5,000 fellowship through the Division of Teaching Excellence and Innovation (DTEI) at UC Irvine.
- Collaborated with faculty mentors to transform INF 43: Introduction to Software Engineering into a project-based course, integrating AI agents and generative AI tools into software engineering education and course projects.
- Completed a 200-hour teaching-development program focused on inclusive pedagogy, accessibility, instructional innovation, and generative AI in education.

Pre-Professor Program (PREPP) Fellow, California State University Aug 2026 – Dec 2026

- Awarded a competitive $1,000 fellowship through the CSU Pre-Professor Program (PREPP).
- Selected to participate in a faculty-preparation program designed to introduce doctoral students to academic careers within the California State University system.
- Engaged in faculty mentorship, professional development, and academic career exploration focused on teaching, research, service, and student success.

Associate Instructor / Lecturer, University of California, Irvine Jun 2023 – Aug 2023

ICS 45J: Programming in Java

- Served as the sole course instructor, responsible for designing the syllabus, assignments, exams, and grading rubrics for an undergraduate course on object-oriented programming in Java.
- Delivered all lectures and lab sessions, providing instruction on core programming concepts, data structures, and software design principles.
- Supervised and coordinated with Teaching Assistants to manage grading, lab support, and student feedback.
- Achieved strong teaching evaluations (Instructor: 8.08/9.00, Course: 8.12/9.00) reflecting effective instruction and student engagement.

Guest Lecturer, University of California, Irvine Feb 2026

SWE 261P: Software Testing (graduate)

- Enhanced course materials by integrating state-of-the-art industry practices and recent research advances in software testing, particularly in mocking, stubbing, and testable design.
- Developed the slides and assignments for mocking/stubbing and testable design.

Guest Lecturer, University of California, Irvine Apr 2025

INF 215: Dynamic Analysis, Testing, and Debugging (graduate)

- Delivered a special lecture on mutation testing and its applications in assessing test adequacy, revealing fault propagation, and guiding test improvement.

Teaching Assistant, University of California, Irvine Jun 2022 – Present

- INF 122: Software Design: Structure and Implementation — Winter 2026
- INF 43: Introduction to Software Engineering — Fall 2021 (Reader), Winter 2022, Fall 2022, Fall 2023, Fall 2025, Spring 2026
- INF 115: Software Testing, Analysis, and Quality Assurance — Spring 2023, Spring 2024, Fall 2024, Spring 2025
- SWE 261P: Software Testing and Debugging — Winter 2023, Winter 2024, Winter 2025

Graduate Research Mentor, University of California, Irvine Oct 2023 – Jun 2024

- Mentored an undergraduate UROP participant in the development of an honors thesis titled "Generative AI for Mutation Testing."
- Advised on research topic selection, proposal development for UROP funding, and the creation of a research poster for the 2024 Undergraduate Research Symposium.
- Conducted weekly research meetings to provide guidance and feedback.

Graduate Research Mentor, University of California, Irvine Sep 2023 – Jun 2025

- Mentored a master's student on the thesis "Automated Detection and Refactoring of Disjoint-Assertion Tangle: A Novel Test Smell."
- Guided the student through experimental design, implementation, paper writing, and thesis development, resulting in a co-authored paper accepted at ASE 2025.

Industry Experience

Financial Analytics Intern, First Capital Securities Nov 2020 – Dec 2020

Supervisor: Ming Shao

- Built data-driven reports and dashboards to support investment decision-making and client advisory services.

Certificates & Training

Essential Class Design Certificate Program, University of California, Irvine — Winter 2026
Mentoring Excellence Certificate Program, University of California, Irvine — Fall 2025
Public Speaking: Activate to Captivate Certificate Program, University of California, Irvine — Winter 2024
Emotional Intelligence Certificate Program, University of California, Irvine — Spring 2024

Professional Affiliations

- Member, IEEE
- Member, Association for Computing Machinery (ACM)
- Member, ACM Special Interest Group on Software Engineering (ACM SIGSOFT)

Awards & Funding

NSF Travel Award for ISSTA — 2026
DTEI Summer Graduate Scholar, University of California, Irvine — 2026
CSU Pre-Professor Program Fellowship — 2026
Undergraduate Research Experience Fellowship (mentor), University of California, Irvine — 2024
Summer Fellowship Award, Department of Informatics, University of California, Irvine — 2024
ICSE Student Mentoring Workshop Award, funded by NSF — 2024
ACM SIGSOFT CAPS Travel Grant — 2023
Chair's Award, University of California, Irvine — 2022
Academic Scholarship, Jilin University, China — 2018, 2019, 2020